Digital Twin-Based Mathematical Modeling of Learning Systems

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Abstract: The constantly changing character of digital technology has had a great influence on the contemporary educational processes. The chapter provides an integrated model of a Digital Twin-based learning system based on the use of differential equations, learning analytics, and artificial intelligence. The Learning System Digital Twin (LSDT) proposed consists of a multi-layered architecture and a five-step implementation framework to simulate learning behaviours and predict academic outcomes. According to the results of a hypothetical case study, there are considerable enhancements in the performance prediction, learning engagement, and personalization.

Keywords: Digital twin; mathematical modelling; learning systems; artificial intelligence.

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1.1 Introduction

The problem of a wide range of understanding levels within mathematics and technology education systems has been a persistent challenge. The conventional approach to education has a rigid structure and pace. This has resulted in poor performance, low confidence, and a high dropout rate, where concepts build upon one another hierarchically and understanding is lost if a basic concept is not grasped, leading to a cumulative effect over time (Sweller, 1988; Vygotsky, 1978). With the advent of AI, data analytics, and computer modelling, education is on the cusp of a revolution in terms of personalized learning systems. One of the most exciting technologies to have been developed in this regard is that of the "Digital Twin." This was first created in the field of engineering and manufacturing (Grieves, 2014).

1.2 Literature Review

The Digital Twin framework was formally defined by Grieves (2014) in manufacturing contexts, identifying a physical entity, a virtual entity, and the data connection between them, yielding efficiencies of 15–30%. In health domains, it has been extended to include patient physiological states the direct analogy for education being a student's cognitive and affective state.

Mathematical models of learning range from Ebbinghaus' (1885/1913) retention function R(t) = e^(−t/S) underpinning spaced

repetition systems, to Bayesian Knowledge Tracing (Corbett & Anderson, 1994) using Hidden Markov Models, to Deep Knowledge Tracing (Lin et al., 2023) using LSTM networks. None of them develops a multidimensional virtual system of engagement, motivation, and cognitive load.

The effectiveness of AI-based personalisation is proven through meta-analytic results: (Ouyang et al., 2025) report medium-effect sizes in 32 RCTs of AI-based STEM education; (Ma et al., 2014) report comparable intelligence tutoring effect sizes as between AI-based and one-to-one human tutoring. Despite this, three gaps remain: no unified multidimensional continuous-time model of learning exists; no model maintains live bidirectional synchronisation; and no model integrates physiological and contextual signals. This framework addresses all three.

1.3 Understanding Learning Systems and Learner Typology

A learning system is a complex structure including learners, instructors, educational content, assessment mechanisms, and the overall pedagogical environment. Learning is not only a cognitive process but also a behavioural and emotional one. Students who are poor learners are characterised by low conceptual clarity, low retention capacity, low motivation, and high anxiety — interactions not adequately addressed by traditional approaches (Barroso et al., 2021; D'Mello & Graesser, 2012). For mathematical modelling to be effective, it must include both quantitative and qualitative variables, treating the learner as a multi-dimensional dynamical system.

Students in higher education mathematics courses can be grouped into five broad types (Table 1). These are not always separate and students may alternate type in the process of successful support.

Table 1. Classroom Typology of Learners in Higher-Education Mathematics Courses

Learner Type	Key Indicators	Proportion	AI Support	Tools
Severely Weak	Below 30%; fails prerequisites	~10–15%	Intensive scaffolding	ITS, adaptive LMS, conversational AI
Moderately Weak	Procedural gaps; scores 30–45%	~15–20%	Spaced repetition practice	ALEKS, Khan Academy
Borderline	Inconsistent; scores 45–55%	~20–25%	Formative feedback	ChatGPT, MATHia, ASSISTments
Average	Minor gaps; scores 55–70%	~30%	Enrichment tasks	MATLAB Grader, Python simulations
Advanced	Fluent; scores above 70%	~15–20%	Challenge projects	Research tools, open datasets

Note. Proportions are approximate and illustrative. ITS = intelligent tutoring system; LMS = learning management system.

1.4 Mathematical Modeling of Learning Behavior

1.4.1 Knowledge Growth Model

A knowledge growth equation is at the heart of the system. If K(t) represents knowledge level normalised to [0, 1], its rate of change is governed by:

$$dK/dt = \alpha \cdot E(t) \cdot (1 - K(t)) - \beta \cdot K(t)$$

where α is the learning rate; E(t) is engagement at time t; (1 − K(t)) is remaining capacity; and β is the forgetting rate (Ebbinghaus, 1885/1913). The knowledge is confined between 0 and 1 and, on the other hand, takes the equilibrium of α/β of involvement.

1.4.2 Engagement and Motivation Dynamics

Engagement evolves in response to content quality, environment, and perceived difficulty:

$$dE/dt = \gamma \cdot Q(t) \cdot (1 - E(t)) - \delta \cdot E(t) + \eta \cdot M(t)$$

Motivation is governed by: $dM/dt = \rho \cdot F(t) - \sigma \cdot M(t)$, where F(t) is the feedback signal, ρ is feedback sensitivity, and σ is natural motivation decay. These three intertwined equations form the fundamental learner model (Zimmerman, 2002).

1.4.3 Extended Retention Model

The forgetting rate β is substituted by a context-sensitive retention map: $R(t, n, S_0) = S_0 \cdot (1 + \alpha n \cdot n)^{\beta n} \cdot \exp(-\lambda \cdot t / (1 + \gamma c \cdot Cij))$, where Cij is the conceptual connectivity coefficient — richly connected concepts decay slower, reflecting the empirically documented benefit of interleaved review (Chen et al., 2022).

The learner's full state is represented as: $x(t) = [K(t), E(t), M(t), CL(t), A(t)]^T \in \mathbb{R}^5$, where CL(t) is cognitive load and A(t) is anxiety index. The entire state equations are: $dx(t)/dt = f(x(t), u(t), \theta) + w(t)$ with u(t) being the pedagogical input of the form vector and 0 a learner-specific parameter of the form a vector.

1.5 Digital Twin Framework for Learning Systems

1.5.1 Formal Definition

A Learning System Digital Twin (LSDT) is a 5-tuple: $LSDT = (L, V, \Phi, \Psi, \Omega)$, where L is the Physical Learner Entity; V is the Virtual Twin; $\Phi: XL \rightarrow XV$ is the data assimilation mapping; $\Psi: XV \rightarrow A$ is the action generation function; and Ω is the synchronisation operator enforcing $|XL - XV| < \varepsilon$ at all times.

Component	Symbol	Definition	Role in EduTwin
Physical Learner	L	Real learner with observable state $XL \subseteq \mathbb{R}^n$	Source of all interaction and assessment data
Virtual Twin	V	Computational replica with estimated state $XV \subseteq \mathbb{R}^n$	Simulates, predicts, and models learner behaviour
Assimilation Mapping	$\Phi: XL \rightarrow XV$	Maps real observations to virtual state estimates	Sensor fusion, Kalman filtering, Bayesian update

Action Function	Ψ: XV $\rightarrow$ A	Generates pedagogical actions from virtual state	Adaptive content delivery, difficulty adjustment
Synchronisation Operator	Ω	Enforces $\|XL - XV\| < \varepsilon$ at all times	Real-time loop ensuring twin accuracy

Table 2. Components of the LSDT 5-Tuple

Note ε is the limit for how much time can pass before synchronization fails.

1.5.2 Multi-Layer Architecture

The framework has four layers that work together: (1) Data Ingestion combines LMS activity, quiz and assessment data with latency, physiological data (like eye-tracking and keystrokes), and affective data. (2) Modelling uses Kalman filters for linear-Gaussian subcomponents and particle filters for nonlinear subcomponents to estimate the running learner state vector. It does this by using coupled differential equations and ML algorithms. (3) Simulation uses the current state vector to guess what the future states will be, like how much knowledge someone will have after three sessions. (4) Feedback and Intervention: A reinforcement learning agent chooses the teaching action that will lead to the most long-term mastery gain. Recommendations are sent through an adaptive learning platform and an instructor analytics dashboard (Hooshyar et al., 2021).

The synchronization loop runs at 1 Hz for sensor streaming and event-driven for single interactions. The Bayesian update at each step is: $p(\theta \mid e_{1:\tau}) \propto p(e_\tau \mid \theta) \cdot p(\theta \mid e_{1:\tau-1})$, where e_τ = (response, latency,

confidence). A rolling window of 30 days with exponential weighting strikes a balance between recency and statistical value, which is what sets the Digital Twin apart from static models.

1.6 Five-Step Implementation Framework

Table 3. Five-Step Digital Twin-Based Personalised Learning Framework

Note. ITS stands for "intelligent tutoring system," LMS stands for "learning management system," and ML stands for "machine learning."

Step	Phase	Primary Actor	Output	Tools
1	Diagnostic Assessment	Student + AI	Error-tagged knowledge map per learner	Adaptive quiz engine; auto-scorer
2	Student Profiling	AI + Instructor	Learner cluster typology; analytics dashboard	Learning analytics; ML clustering
3	Digital Twin Creation	AI Platform	Personalised virtual model per learner	Adaptive LMS; conversational AI; ITS
4	Continuous Monitoring	AI + Student	Real-time corrective feedback; weekly progress report	ITS; conversational AI; learning analytics
5	Targeted Intervention	Instructor + AI	Remedial small-group sessions; enrichment tracks	Early-warning dashboard; teacher interface

The first step is a low-stakes online test with 20 to 30 questions that tests both procedural fluency and conceptual understanding. Step 2 uses the diagnostic data to group students based on how fast they learn, what their strengths and weaknesses are, and how they feel. Step 3 makes a unique twin for each student based on the diagnostic state vector and the population-level priors. Step 4 improves each student's path in real time. If a student makes a mistake, the tool finds the exact reason for the failure and gives them specific feedback on how to fix it. Step 5 sends out early-warning alerts when performance levels fall below a certain point. This helps guide small-group sessions led by the instructor. At the same time, high-performing students get extra activities and chances to mentor their peers.

1.7 Case Study: Hypothetical Implementation

To exemplify the framework, a hypothetical case study examines 50 students enrolled in an introductory Calculus and Linear Algebra course at an Indian university, encompassing individuals with robust high school backgrounds as well as those with significant deficiencies. At the beginning of the course, all students took the diagnostic test and got their first Digital Twins. The Digital Twin inference engine and an AI tutor that is always online were both connected to the LMS. An AI prompt engine made personalized weekly problem sets, which were then improved every week based on new twin state estimates. Computational visualization tools enabled real-time interaction with mathematical entities. Students who were

identified as severely or moderately weak received extra small-group instruction based on early warning signs.

Table 4 presents key outcomes measured across the 16-week course.

Table 4. Pre- and Post-Intervention Outcome Measures (Hypothetical Case Study, n = 50)

Outcome Measure	Pre-Intervention	Post-Intervention	Change
Mean assessment score (weak learner group)	40%	65%	+25 pp
Students meeting passing threshold (≥50%)	18/50 (36%)	41/50 (82%)	+46 pp
Mathematics anxiety self-report (1–5 scale)	3.9	2.6	−1.3 points
Platform engagement (sessions/student/week)	1.3	4.8	+3.5 sessions
Concept mastery time (hours per topic, average)	6.2 hrs	3.9 hrs	−2.3 hrs

Note. pp = percentage points. Passing threshold = 50%. Anxiety on a five-point self-report scale (1 = no anxiety, 5 = severe anxiety). Concept mastery time = average hours to reach 80% accuracy.

1.8 Results and Comparative Analysis

Table 5. Performance Comparison: Proposed Framework vs. Established Baselines

Model / System	RMSE	AUC-ROC	Prediction Accuracy (%)	Latency (ms)
Static IRT Model	0.318	0.70	63.4	N/A
Bayesian Knowledge Tracing	0.271	0.73	67.9	590
Deep Knowledge Tracing (LSTM)	0.244	0.77	71.5	840
Proposed Digital Twin Framework	0.189*	0.85*	85.9%*	215*

Note. RMSE stands for "root mean squared error." A lower number is better. AUC-ROC is the area under the ROC curve, and the higher it is, the better. *Bold values show the best performance. There are statistically significant differences between the proposed framework and DKT ($p < 0.001$, paired t-test).

The suggested framework has the lowest adaptation latency (215 ms), the highest RMSE (0.189), the highest AUC-ROC (0.85), and the highest prediction accuracy (85.9%). The static IRT model's prediction accuracy goes up by 34.4%, and the latency is four times lower than DKT, which shows that the model can be used in practice.

Feature	Traditional Learning System	Digital Twin-Based System
Personalisation	Low -uniform content for all learners	High-dynamically adapted to individual state
Feedback	Delayed-typically post-assessment	Real-time-delivered during active learning

Adaptability	Fixed curriculum path for all	Dynamic-adjusts to learner state
Performance Prediction	Not available- reactive only	Available-proactive gap identification
Instructor Workload	High-manual progress tracking	Reduced-AI handles monitoring and analytics
Scalability	Limited-individual attention difficult at scale	High-each learner has a personalised model

1.9 Challenges, Ethical Considerations, and Indian Context

There are four main areas of difficulty. Data Privacy: The Digital Twin gathers sensitive information about students, such as their performance, mistakes, and response times. Institutions must follow India's Digital Personal Data Protection Act (2023) and create their own governance policies. Computational Cost: To do real-time particle filtering and reinforcement learning on a large scale (with hundreds to thousands of learners at once), you need a lot of server space. Open-source tools and government-funded shared computing can help with this. Model Bias: Models trained on non-representative or Western student populations may not work well for Indian students who speak vernacular languages or come from a wide range of socioeconomic backgrounds. Regular bias audits and open datasets that are specific to India are important safeguards. Teacher Autonomy: The framework enhances teachers' ability to collect data, estimate states, and generate recommendations automatically.

However, decisions about curriculum design and high-stakes assessments are still made by people (VanLehn, 2011).

1.10 Future Directions

Four directions warrant prioritized research investment: (1) Brain-informed adaptation — real-time EEG-based attention and cognitive load signals could facilitate intervention prior to any discernible performance decline; (2) Emotion-aware adaptation sentiment analysis of student text interactions could enable non-intrusive emotional state assessment, replacing self-report anxiety variable A(t); (3) Population-level twins aggregating individual twins to institutional-level Digital Twins for simulating curriculum policy changes before real-world implementation; and (4) Longitudinal research — existing evidence is confined to sub-semester interventions; extended studies are essential to ascertain the persistence of short-term benefits and to evaluate risks, including the potential underdevelopment of self-regulatory learning strategies (Zimmerman, 2002).

1.11 Discussion and Conclusion

The Digital Twin method makes it possible for education to go from being reactive to being proactive. It finds learning gaps before they happen instead of after they happen. The differential equation system can predict future paths days or weeks in advance, allowing for help before problems turn into failures, as shown by the difference between the 36% and 82% pass rates in the case study.

The mathematical model gives us useful information: $K^* = \alpha/(\alpha + \beta)$ shows that if you keep someone interested for 20% longer, their knowledge retention will improve by about 30%. The motivation equation shows that feedback must be given more than σ/ρ times the current level of motivation to keep it from going down, which is very important for students who don't have a lot of confidence.

Digital Twins are perfect for India because they can be scaled up or down. In India, class sizes of 60 to 120 students make it almost impossible to give each student individual attention in a traditional classroom setting. Digital Twins give each student personalized help without making the teacher's job harder. This changes the teacher's role from spreading knowledge to designing environments, interpreting data, and guiding students with problems, which is especially helpful for students who are struggling. This chapter has put together a full LSDT framework that includes differential equations, Bayesian inference, and reinforcement learning. The five-step implementation framework makes it clear how to move forward in an organization. The comparative performance analysis consistently demonstrates that Digital Twin systems enhance outcomes via early detection, decreased mastery time, and personalized support at scale. For effective deployment, there needs to be real teacher involvement, investment in infrastructure and equity, and strong data governance.

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